

M3D RC Delay

新版 Presentation 演讲思路与中英对照演讲稿

Based on the revised 12-slide presentation: M3D_RC_Delay(1).pptx

目标：约 10 分钟。按新版 PPT 顺序逐页讲，强调“零背景听众也能听懂”。

整场主线

Definitions → Why M3D → Real device examples → T/R → R & C → Inverter → Delay measurement → Literature evidence → 1T1R test → Ring oscillator + SPICE

最后始终回到一个问题：

When is $t_{M3D} < t_{2D}$?

Slide 1 — Monolithic 3D Integration: RC Delay of Vertically Integrated Circuits

建议时间：约 30 秒

演讲思路

只说研究对象和最终问题，不展开术语。

中文演讲稿

大家好，今天我介绍的题目是单片三维集成中的 RC delay。

这个项目主要研究上下堆叠的 TFT circuit、它们之间的垂直电连接，以及 MAC 作为 inter-tier dielectric 时对电路速度的影响。

整个 presentation 最终想回答一个问题：在我们的 stacked TFT platform 里，什么时候 M3D 的 propagation delay 会比 2D reference 更小？

English Script

Good morning everyone. Today I will present my project on RC delay in monolithic 3D integrated circuits.

The project focuses on vertically stacked TFT circuits, the electrical connection between the tiers, and the possible role of MAC as the inter-tier dielectric.

The final question is: under what conditions can the propagation delay of our stacked M3D TFT platform become lower than that of a matched 2D reference?

过渡 / Transition

中文：在进入物理之前，我先给出今天的 presentation structure。

English: Before going into the physics, I will first show the structure of the presentation.

Slide 2 — Presentation structure

建议时间：约 25 秒

演讲思路

对应 mentor 要求：让没有背景的听众先知道你要讲什么。

中文演讲稿

今天我分成四部分。

第一部分是 motivation：为什么 interconnect RC delay 很重要，以及 2D 和 M3D 有什么区别。

第二部分是 device and link：我会解释 driver、receiver、MIV，以及 resistance 和 capacitance 在哪里产生。

第三部分是 measurement：包括 inverter delay、1T1R 的 R/C 测量，以及 ring oscillator。

最后是 decision：用 matched 2D/M3D controls 和 SPICE 来判断 M3D 是否真的更快。

English Script

The presentation has four parts.

First is the motivation: why interconnect RC delay matters, and how 2D and M3D differ.

Second is the device and link section: I will define the driver, receiver and MIV, and show where the resistance and capacitance come from.

Third is measurement: inverter delay, 1T1R resistance and capacitance measurements, and the ring oscillator.

Finally, I will discuss how matched 2D and M3D controls together with SPICE can be used to decide whether M3D is actually faster.

过渡 / Transition

中文：先把几个后面会反复出现的术语定义清楚。

English: I will first define several terms that will appear throughout the presentation.

Slide 3 — Key terms before the physics

建议时间：约 50 秒

演讲思路

这页是为了防止听众后面被 driver/receiver/MIV/t2D/tM3D 卡住。

中文演讲稿

先定义几个术语。

Driver 是前一级电路，也就是负责产生并发送信号的一端。

Receiver 是后一级电路，它接收这个信号，同时也会给前一级带来 electrical load。

中间的 MIV 是 Monolithic Inter-tier Via，也就是连接上下两个 tier 的垂直电连接。

这里的 t2D 表示 planar 2D reference 的 propagation delay，tM3D 表示 stacked M3D circuit 的 propagation delay。

另外，Pdyn 是 dynamic switching power。这里的 driver 和 receiver 只是信号链里的角色，不是特殊的器件名称。整个 inverter 都可以作为一个 driver stage，另一个 inverter 也可以作为 receiver stage。

English Script

Let me first define the key terms.

The driver is the previous circuit stage that generates and sends the signal.

The receiver is the next circuit stage that receives the signal and also acts as an electrical load.

The MIV, or monolithic inter-tier via, is the vertical electrical connection between the two tiers.

Here, t2D means the propagation delay of the planar 2D reference, while tM3D means the delay of the stacked M3D circuit.

Pdyn means dynamic switching power. Driver and receiver are roles in the signal path, not special device names. A complete inverter can act as the driver stage, and another inverter can act as the receiver stage.

过渡 / Transition

中文：有了这些定义，就可以看为什么 M3D 可能降低 delay。

English: With these definitions in place, we can now see why M3D may reduce delay.

Slide 4 — Why M3D? Interconnect delay is the bottleneck

建议时间：约1 分钟

$$R = \rho L / A$$

$$C \approx \epsilon_0 k A / d$$

演讲思路

核心：M3D 不是“3D 就一定快”，而是“shorter link vs new parasitics”的竞争。

中文演讲稿

在传统 2D routing 中，driver 和 receiver 主要通过同一层里的横向金属线连接。

真实金属线有电阻。根据 $R = \rho L / A$ ，长度 L 越大，wire resistance 通常越大。

互连也会和周围导体形成寄生电容。简单地看，C 大约与 k 和 overlap area 成正比，与距离 d 成反比。

M3D 的机会是把一部分较长的 horizontal wire 变成很短的 vertical link，也就是 MIV。

这样可以降低 interconnect length，但同时又会增加 contact、via、fringing 和 coupling parasitics。

因此 M3D 真正胜出的条件是：shorter interconnect 带来的 benefit，大于新增 vertical parasitics 带来的 penalty。

English Script

In conventional 2D routing, the driver and receiver are mainly connected by lateral metal wires within the same tier.

A real metal wire has resistance. From $R = \rho L / A$, a longer interconnect generally gives a larger wire resistance.

The interconnect also forms parasitic capacitance with nearby conductors. In a simple picture, capacitance increases with dielectric constant and overlap area and decreases with separation.

The opportunity in M3D is to replace some long horizontal wires with a very short vertical link, or MIV.

This reduces the interconnect length, but the stacked structure also introduces new contact, via, fringing and coupling parasitics.

Therefore, M3D wins only when the benefit of shorter interconnects is larger than the penalty from the additional vertical parasitics.

过渡 / Transition

中文：接下来先看一些真实的 inverter 和 stacked-device 例子，建立器件直觉。

English: Next, I will show some real inverter and stacked-device examples to build device-level intuition.

Slide 5 — What can a vertically integrated circuit look like?

建议时间：约 50 秒

演讲思路

不要深入讲材料细节；只让听众知道项目不是抽象方框，真实器件大概长什么样。

中文演讲稿

这里给出几个真实 device example。

左边是 MoTe₂/MoS₂ complementary inverter，它展示了 2D semiconductor transistor 如何组成 CMOS-like circuit。

中间是 MoTe₂/IGZO hybrid CMOS，它把 2D material 和 oxide TFT 结合起来。

右边是 MAC 的结构示意。MAC 是 disordered carbon network，在我们的项目里它的角色是 candidate inter-tier dielectric。

因此我们项目的目标不是只研究一种材料，而是把 stacked TFT、MAC inter-tier dielectric 和真实的 vertical-circuit delay measurement 连在一起。

English Script

Here I show several real device examples.

On the left is a MoTe₂/MoS₂ complementary inverter, showing how two different semiconductor transistors can form a CMOS-like circuit.

In the middle is a MoTe₂/IGZO hybrid CMOS inverter combining a 2D semiconductor with an oxide TFT.

On the right is the disordered carbon structure of MAC. In our project, MAC is considered as a candidate inter-tier dielectric.

So the project is not only about one material. The target is to connect stacked TFTs, the MAC inter-tier dielectric and an actual measurement of vertical-circuit delay.

过渡 / Transition

中文：为了系统地描述前一级和后一级之间的连接，我用 T/R notation。

English: To describe the connection between the previous and next stages systematically, I use the T/R notation.

Slide 6 — T/R structures: count the signal source and the load

建议时间：约1 分钟

More receivers → Cload ↑ → delay ↑

演讲思路

强调：这是 signal-transmission test structure，不是 memory cell。

中文演讲稿

这里的 T 代表 transmitter 或 driver，R 代表 receiver 或 load。

这些结构的作用是传输一个 voltage transition，并研究 vertical link 如何加载和减慢这个信号；它们不是 memory cell。

1T0R 表示只看 driver，0T1R 表示只看 receiver。

1T1R 是一个 driver 驱动一个 receiver，也就是最简单的 point-to-point link。

1T2R 表示一个 driver 同时驱动两个 receiver，因此 fan-out 更大。

从电学上，transmitter 有有限的 output resistance R_{drv} ，而 receiver 主要表现为 input capacitance C_{in} 。

所以 receiver 数量增加时，总 C_{load} 增加，delay 通常也会增加。

English Script

Here, T means transmitter or driver, and R means receiver or load.

These are signal-transmission test structures. Their role is to transmit a voltage transition and study how the vertical link loads and slows the signal; they are not memory cells.

1T0R means driver only, while 0T1R means receiver only.

1T1R is one driver connected to one receiver, so it is the simplest point-to-point link.

1T2R means that one driver drives two receivers, which increases the fan-out.

Electrically, the transmitter has a finite output resistance, R_{drv} , while the receiver mainly contributes an input capacitance, C_{in} .

Therefore, increasing the number of receivers increases the total load capacitance and normally increases delay.

过渡 / Transition

中文：接下来把这个 vertical link 里面真正的 R 和 C 标出来。

English: Next, I will identify the actual resistance and capacitance inside this vertical link.

Slide 7 — Where exactly do R and C come from?

建议时间：约 1.1 分钟

$$R_{wire} = \rho L / A = R_s \cdot L / W$$

$$C \approx \epsilon_0 k A_{overlap} / d$$

$$\tau E = R_{drv}(C_{out} + C_{int} + C_L) + R_{int}(C_{int}/2 + C_L)$$

演讲思路

一边看 stacked cross-section，一边看 π -RC model。

中文演讲稿

在 stacked cross-section 里，R 和 C 都有明确的物理位置。

首先是电阻。driver 自己有 output resistance R_{drv} 。然后信号经过 semiconductor contact、vertical via 和 metal line，这些可以合并成 R_{int} 。

其中 wire resistance 可以写成 $R = \rho L / A$ ，或者用 sheet resistance 写成 R_s 乘 L / W 。

电容来自相邻导体之间。上下层金属隔着 MAC 或其他 ILD 会形成 inter-tier capacitance；边缘电场形成 fringing capacitance；receiver gate 和 measurement setup 还会提供 load capacitance。

为了先建立一个简单模型，可以把 vertical link 表示成 lumped pi-RC network。

这里的 Elmore time expression 只是一个第一阶 delay scale。真正的 50% propagation delay 最后仍然从 transient waveform 里面提取。

English Script

In the stacked cross-section, the resistance and capacitance have clear physical locations.

First, the driver itself has an output resistance, R_{drv} . The signal then passes through the semiconductor contact, the vertical via and the metal line, which can be combined into R_{int} .

The wire resistance can be written as $\rho L / A$, or using sheet resistance as $R_s \text{ times } L / W$.

The capacitance comes from neighboring conductors. The top and bottom metals separated by MAC or another inter-layer dielectric form inter-tier capacitance. Edge electric fields create fringing capacitance, while the receiver gate and the measurement setup contribute load capacitance.

As a first model, the vertical link can be represented by a lumped pi-RC network.

The Elmore expression shown here is only a first-order delay scale. The actual 50-percent propagation delay is still extracted from the transient waveform.

过渡 / Transition

中文：有了 link model 后，还必须知道 driver 本身够不够强，所以接下来讲 inverter。

English: Once we have the link model, we also need to know how strong the driver is, so next I will discuss the inverter.

Slide 8 — Inverter: the basic logic test circuit

建议时间：约 1 分钟

$$Y = \text{NOT}(A)$$

$$V_{IN} = 0 \rightarrow V_{OUT} \approx V_{DD}$$

$$V_{IN} = V_{DD} \rightarrow V_{OUT} \approx 0$$

$$t \approx C_{load} \cdot V_{DD} / I_{eff}$$

演讲思路

回答 mentor：inverter 能做什么？必须哪些元件？测什么曲线？

中文演讲稿

Inverter 是最基础的 logic test circuit。

标准 CMOS inverter 由一个 p-FET 和一个 n-FET 组成。输入低时，p-FET pull-up，所以输出接近 V_{DD} ；输入高时，n-FET pull-down，所以输出接近 0。

因此它实现的逻辑就是 NOT。

也可以做 resistor-loaded inverter，也就是一个 transistor 加一个 resistor，但性能通常不如 complementary CMOS。

而 transistor 加 capacitor 本身不是一个完整的 static inverter，capacitor 更像 dynamic load 或 storage element。

对于 transistor，我们主要测 ID-VG 和 ID-VD；对于 inverter，我们测 VOUT-VIN 的 VTC，以及 VIN(t)、VOUT(t) 的 transient waveform。

drive current 越大，给相同 Cload 充放电越快，所以 t 大约和 Cload·VDD/Ieff 成正比。

English Script

The inverter is the most basic logic test circuit.

A standard CMOS inverter uses one p-FET and one n-FET. When the input is low, the p-FET pulls the output up toward VDD. When the input is high, the n-FET pulls the output down toward zero.

So the logic function is simply NOT.

A resistor-loaded inverter, using one transistor and one resistor, is also possible, although it is generally less ideal than complementary CMOS.

A transistor-capacitor pair alone is not a complete static inverter; the capacitor mainly acts as a dynamic load or storage element.

For the individual transistors, we measure ID-VG and ID-VD. For the inverter, we measure the VOUT-VIN transfer characteristic and the transient VIN and VOUT waveforms.

A stronger drive current charges or discharges the same load capacitance more quickly, so the switching time is roughly proportional to Cload times VDD divided by Ieff.

过渡 / Transition

中文：接下来就在 transient waveform 上直接定义 propagation delay 和 edge time。

English: Next, I will define propagation delay and edge time directly on the transient waveform.

Slide 9 — How to measure propagation delay and edge time

建议时间：约 1 分钟

$$tp_{HL} = 50\% \text{ VIN} \uparrow \rightarrow 50\% \text{ VOUT} \downarrow$$

$$tp_{LH} = 50\% \text{ VIN} \downarrow \rightarrow 50\% \text{ VOUT} \uparrow$$

$$tp = \frac{1}{2}(tp_{HL} + tp_{LH})$$

$$\text{single-pole RC: } \tau = RC, \quad t_{50} \approx 0.69RC$$

$$10\text{--}90\% \text{ edge: } tr, tf \approx 2.2RC$$

演讲思路

本页只说“怎么从图上量”，不要推导。

中文演讲稿

实际测量时，我们给 inverter 一个 pulse input，同时用 oscilloscope 记录 VIN 和 VOUT。

如果 VIN 上升、VOUT 下降，我们从两个 50% crossing 之间的时间差得到 tpHL。

反过来，VIN 下降、VOUT 上升，得到 tpLH。

平均 propagation delay 是两者的平均值。

对于一个简单的一阶 RC response，time constant 是 RC，而 50% response time 大约是 $0.69RC$ 。

rise time 和 fall time 一般用 10% 到 90% 或 90% 到 10% 来定义。

除了 delay，还要检查 voltage swing、crosstalk、noise、droop 和 waveform distortion，因为信号不仅要快，也必须保持逻辑有效。

English Script

In the actual measurement, we apply a pulse input to the inverter and use an oscilloscope to record both VIN and VOUT.

When VIN rises and VOUT falls, the time difference between the two 50-percent crossings gives tpHL.

For the opposite transition, VIN falls and VOUT rises, giving tpLH.

The average propagation delay is the average of these two values.

For a simple first-order RC response, the time constant is RC, and the 50-percent response time is approximately $0.69RC$.

Rise time and fall time are usually defined using the 10-to-90-percent and 90-to-10-percent transitions.

In addition to delay, we should also check voltage swing, crosstalk, noise, droop and waveform distortion, because the signal must remain logically valid as well as fast.

过渡 / Transition

中文：文献里已经可以看到，真实器件的 parasitic structure 会明显改变 dynamic delay。

English: The literature already shows that parasitic device structures can strongly change the dynamic delay.

Slide 10 — Literature evidence: real devices show parasitic-limited speed

建议时间：约 1.1 分钟

演讲思路

一定强调：不是速度排名，而是学习“什么 parasitic 要测”。

中文演讲稿

这里是几个实际 inverter 的动态结果。

在同一篇 MoTe2/MoS2 inverter 论文里，作者在相同 2 V、1 kHz 条件下比较了两种结构。

SiO₂/p+-Si substrate 的结构因为 overlap 更大，delay 大约是 480 微秒；glass structure 的 overlap 更小，动态响应明显更快。

同一工作中还报告了大约 25 微秒的 inverter switching delay，不过这个数对应不同 bias condition，所以不能和 480 微秒直接做简单速度排名。

另一篇 MoTe2/IGZO hybrid CMOS 在 1 V、1 kHz 下报告了大约 200 微秒的 RC delay，并指出 contact resistance 和 overlap capacitance 是重要来源。

因此这些文献最重要的启示不是哪个材料最快，而是：必须把 geometry、contact R 和 parasitic C 实际测出来。

English Script

Here are several experimental dynamic results from real inverters.

In the same MoTe2/MoS2 inverter paper, the authors compared two structures under the same 2-volt, 1-kilohertz condition.

The SiO₂/p-plus-Si structure had a larger overlap region and showed a delay of about 480 microseconds, while the glass structure had a smaller overlap and a much faster dynamic response.

The same work also reported an inverter switching delay of about 25 microseconds, but under a different bias condition, so these values should not be treated as a simple speed ranking.

A MoTe2/IGZO hybrid CMOS inverter reported about 200 microseconds of RC delay at 1 volt and 1 kilohertz, and identified contact resistance and overlap capacitance as important parasitic sources.

The key lesson is therefore not which material is fastest, but that geometry, contact resistance and parasitic capacitance must be measured explicitly.

过渡 / Transition

中文：基于这些文献，我下一步的 test vehicle 应该先把 R、C 和总 delay 分开测。

English: Based on these results, the next test vehicle should measure R, C and total delay as independently as possible.

Slide 11 — Proposed 1T1R test vehicle: measure R, C and delay independently

建议时间：约 1.2 分钟

$$R = V / I$$

演讲思路

这页是研究计划的核心：结构、R measurement、C measurement、delay measurement、controls。

中文演讲稿

我提出的 starting test vehicle 是一个简单 1T1R link。

左边是一个 inverter 或 TFT driver，中间通过 MIV 和 MAC vertical link，右边只接一个 known receiver gate。

同时制作一个 matched planar reference。

R 的测量可以独立进行：force current、sense voltage，然后得到 $R = V/I$ ；更严格时可以使用 Kelvin 或其他专门结构减少 contact ambiguity。

C 的测量则使用 top electrode、MAC/ILD、bottom electrode 形成的 capacitor structure，通过 impedance 或 C-V 测量得到 $C(f,V)$ 。

delay measurement 使用 pulse generator 和两路 oscilloscope channel，同时记录 $V_{IN}(t)$ 和 $V_{OUT}(t)$ ，提取 t_p 、 t_r 、 t_f 和 voltage swing。

为了公平比较，VDD、fan-out、probe load、temperature 都要保持一致，同时需要 de-embed pads 和 probes。

English Script

My proposed starting test vehicle is a simple 1T1R link.

On the left is an inverter or TFT driver. The signal passes through the MIV and MAC vertical link, and on the right there is one receiver gate with a known input capacitance.

A matched planar reference should be fabricated at the same time.

The resistance can be measured independently by forcing current and sensing voltage, giving R equals V over I . More rigorous Kelvin-type structures can also be used to reduce ambiguity from contact resistance.

For capacitance, a top electrode, MAC or other inter-layer dielectric, and a bottom electrode form a capacitor structure. Impedance or C-V measurements then provide C as a function of frequency and voltage.

For delay, a pulse generator and two oscilloscope channels record V_{IN} of time and V_{OUT} of time, from

which t_p , t_r , t_f and voltage swing are extracted.

For a fair comparison, VDD, fan-out, probe load and temperature must be matched, and the pad and probe parasitics should be de-embedded.

过渡 / Transition

中文：最后再把多个 stage 连起来，用 ring oscillator 做 circuit-level measurement，并用 SPICE 解释结果。

English: Finally, multiple stages can be connected into a ring oscillator for circuit-level measurement, and SPICE can be used to explain the result.

Slide 12 — Ring oscillator + SPICE: the circuit-level closure

建议时间：约 1.2 分钟

$$N = 3, 5, 7, \dots$$

$$t_{p,stage} \approx 1 / (2Nf_{osc})$$

$$3 \text{ CMOS stages} \times 2 \text{ TFTs} = 6 \text{ TFTs}$$

演讲思路

先回答 ring oscillator，再讲 SPICE，最后回到核心问题。

中文演讲稿

最后是 circuit-level closure。

Ring oscillator 需要奇数个 inverter stage，例如 3、5 或 7 个。

最简单的 3-stage CMOS ring oscillator 有 3 个 inverter；如果每个 inverter 是 p-FET 加 n-FET，那么总共是 6 个 TFT，再根据 stacked architecture 加入相应的 vertical links。

通过测 oscillation frequency，可以估算平均 stage delay：

$t_{p,stage}$ 大约等于 1 除以 $2Nf_{osc}$ 。

然后把前面测到的 TFT characteristics、 R_{int} 、 C_{int} 和 load 放进 SPICE。

SPICE 可以模拟 VIN 和 VOUT transient，并在 matched conditions 下比较 2D 和 M3D。

这样我们就能一步一步区分：transistor 本身造成多少 delay，contact 和 vertical link 又增加多少 delay，MAC 通过改变 inter-tier capacitance 能不能带来实际改善。

最后回到核心问题：M3D 只有在 wire savings 带来的 delay reduction 大于新增 parasitic delay 时，才真正胜过 2D。

English Script

Finally, we close the analysis at the circuit level.

A ring oscillator requires an odd number of inverter stages, for example three, five or seven.

The simplest three-stage CMOS ring oscillator contains three inverters. If each inverter uses one p-FET and one n-FET, the circuit contains six TFTs, plus the required vertical links in the stacked architecture.

By measuring the oscillation frequency, the average stage delay can be estimated from $t_{p,stage}$ approximately equal to one over two N_{fosc} .

The measured TFT characteristics, R_{int} , C_{int} and load can then be inserted into a SPICE model.

SPICE can simulate the VIN and VOUT transients and compare matched 2D and M3D circuits.

This allows us to separate the delay contributions step by step: the transistor baseline, the contact and vertical-link penalty, and the possible improvement from changing the inter-tier capacitance with MAC.

The final question is therefore unchanged: M3D is faster only when the delay reduction from wire savings is larger than the additional parasitic delay.

Closing / 最后 15 秒

中文结束语

总的来说，我的研究路线是先把术语和物理结构讲清楚，再分别测 transistor、R、C 和 transient delay，最后用 ring oscillator 和 SPICE 做 circuit-level validation。

真正的目标不是单独证明 MAC 是 low-k，而是证明在 matched conditions 下，它是否能帮助我们的 stacked TFT platform 得到更低的 propagation delay。谢谢大家。

English Closing

In summary, the plan is to first define the device structure clearly, then measure the transistor behavior, resistance, capacitance and transient delay independently, and finally validate the circuit-level behavior with a ring oscillator and SPICE.

The real goal is not simply to show that MAC is low-k, but to determine whether it can help our stacked TFT platform achieve a lower propagation delay under matched conditions. Thank you.

最后只背这一条逻辑

Define → Compare 2D/M3D → Locate R/C → Use inverter → Measure delay → Validate with literature → Build 1T1R → Ring oscillator + SPICE → $t_{M3D} < t_{2D}$?